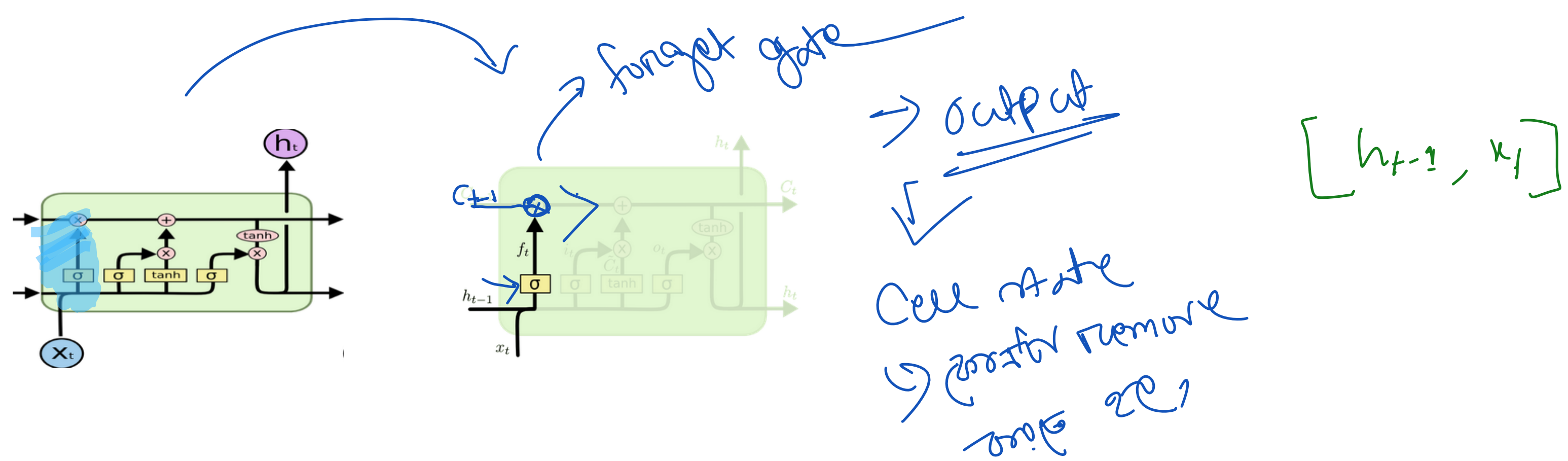




input:

- ①  $C_{t-1}$  → previous cell state
- ②  $h_{t-1}$  → previous hidden state
- ③  $x_t$  → current word / current input

over

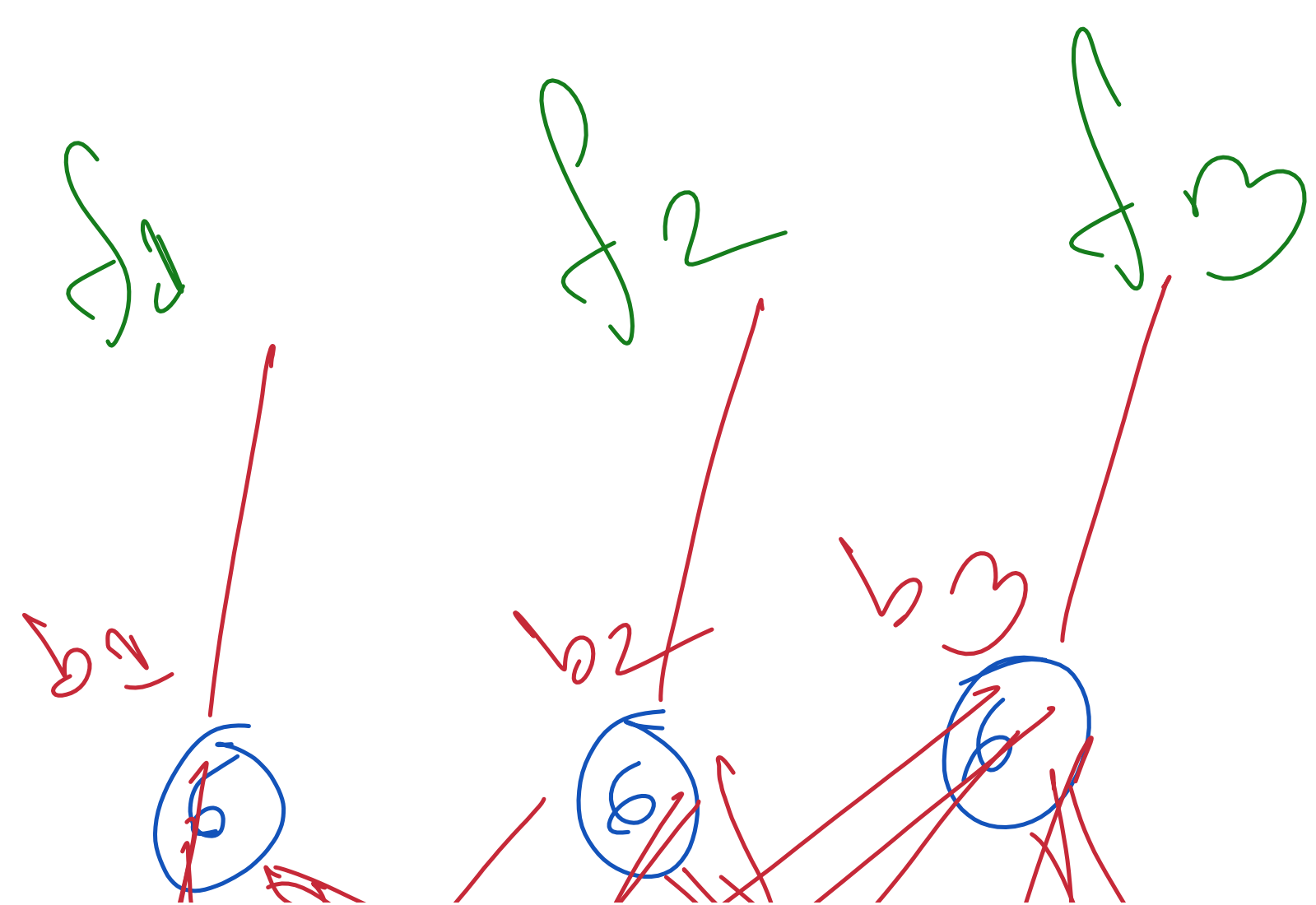
- ①  $f_t$  calculate over
- ②  $C_{t-1}$  ①  $f_t$  calculate over

~~neural network layer~~ → node = 3

→  $C_{t-1}$  → 3

→  $h_{t-1}$  → 3

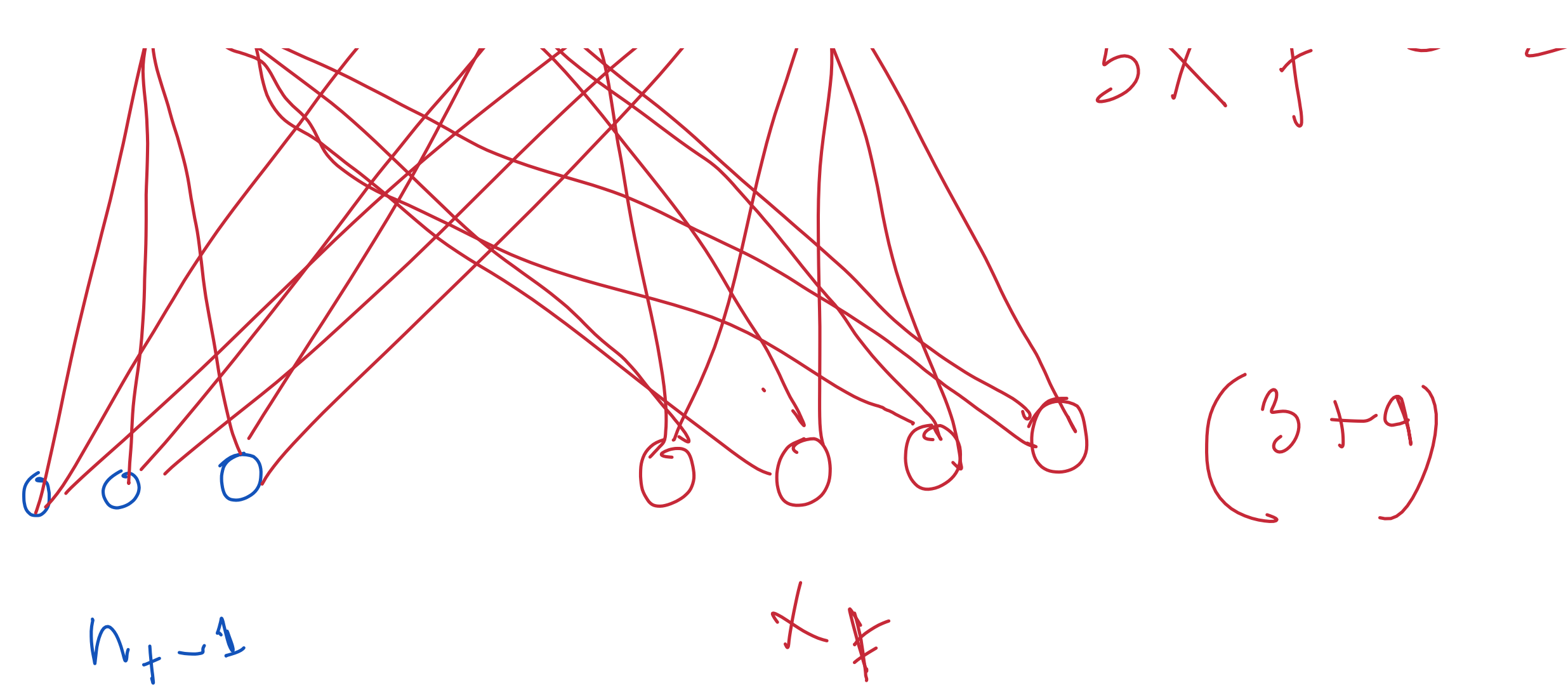
→  $x_t$  →  $[x_{i1} \ x_{i2} \ x_{i3} \ x_{i4}]$



→  $[f_1 \ f_2 \ f_3] \rightarrow f_t(3)$

... = 9-1 weight





Equation form:

$$x_t = [n_{t-1}, x_t] \cdot w_f + b_f$$

$$f_t = \sigma(w_f \cdot [n_{t-1}, x_t] + b_f)$$

$3 \times 7$

→  $(3 \times 7) \oplus (7 \times 1) \rightarrow (7 \times 1)$



$$\cancel{A} C_{t-1} \oplus f_t$$

$$C_{t-1} = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$$

$$f_t = \begin{bmatrix} \uparrow \frac{1}{2} & \uparrow \frac{1}{2} & \uparrow \frac{1}{2} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 2.5 & 3 \end{bmatrix}$$

$$C_{t-1} \oplus f_t$$

remove



$$O_{+1} = [4, 5, 6] \quad \text{371}$$

$$f_{+2} = [1, 2, 1] \quad \text{371}$$

- (371)